

MMS source terms for the production PNP–BV stack

logc_muh + Cs⁺/SO₄²⁻ multi-ion + Stern Robin

PNPInverse / Forward solver verification

May 27, 2026

This is the one-page version. Full algebra, clip-safety bounds, runtime asserts, and convergence results live in:

- derivation — docs/solver/mms_pnpbv_muh_multi_ion_stern_derivation.md
- full writeup — writeups/May13th/mms_source_terms_derivation.tex
- source builder — scripts/verification/mms_pnpbv_muh_multi_ion_stern.py
- test — tests/test_mms_logc_muh_multi_ion_stern.py

1 The stack under test

The production “no-speculative-physics” stack is the one exercised by the demo solver_demo_slide15_no_speculative_cs.py. The mixed space is $W = V_{O_2} \times V_{H_2O_2} \times V_{\mu_H} \times V_\varphi$ with

$$U = (u_{O_2}, u_{H_2O_2}, \mu_H, \varphi), \quad z = (0, 0, +1, \cdot), \quad e_m z_H = +1.$$

The proton primary is its electrochemical potential μ_H , so concentrations are reconstructed as $c_H = \exp(\mu_H - e_m z_H \varphi)$ and $c_i = \exp(u_i)$ for $i \in \{O_2, H_2O_2\}$ (**muh transform**). Domain $\Omega = (0, 1)^2$, electrode at $y = 0$, bulk at $y = 1$. Two analytic Bikerman counterions $\{Cs^+, SO_4^{2-}\}$ enter through the shared- θ closure, giving the steric potential $\mu_{ster} = -\ln \theta_{inner}$ and per-ion charges $c_k^{ster}(\varphi; c_{dyn})$; parallel R2e/R4e Butler–Volmer in log-rate form gives the surface rates R_j . A Stern Robin BC replaces the φ -Dirichlet at the electrode.

2 Manufactured solution

Smooth fields that meet all bulk Dirichlets exactly ($\delta_i = 0.30$, $\alpha_0 = \alpha_1 = \gamma = 0.5$):

$$\begin{aligned} c_i^{ex}(x, y) &= c_{0,i} [1 + \delta_i \cos(\pi x) (1 - y)^2], \quad i \in \{O_2, H_2O_2, H\}, \\ \varphi^{ex}(x, y) &= (1 - y) [\alpha_0 + \alpha_1 \cos(\pi x)] + \gamma y (1 - y) \cos(\pi x), \\ \mu_H^{ex} &= \ln c_H^{ex} + e_m z_H \varphi^{ex}. \end{aligned}$$

At $y = 1$: $c_i^{ex} = c_{0,i}$, $\varphi^{ex} = 0$ (bulk Dirichlets met). At $y = 0$: $\varphi^{ex}(x, 0) = \alpha_0 + \alpha_1 \cos(\pi x) \in [0, 1] \neq \varphi_{app}^{model} \approx 21.4$, so the Stern Robin is non-trivially driven. At $x \in \{0, 1\}$: $\partial_x(\cdot) = 0$ for every field, so the side walls are auto zero-flux and need no source. All closure quantities (A_{dyn}^{ex} , D^{ex} , $c_k^{ster,ex}$, μ_{ster}^{ex} , R_j^{ex}) and the mobility-convention fluxes $J_i^{ex} = D_i c_i^{ex} (\nabla u_i^{ex} + \nabla \mu_{ster}^{ex})$, $J_H^{ex} = D_H c_H^{ex} (\nabla \mu_H^{ex} + \nabla \mu_{ster}^{ex})$ are then exact UFL expressions in (x, y) .

3 The four source terms and how they plug in

Mechanism. The residual is assembled byte-faithfully by production. For each weak-form contribution $+\int X v d\mu$ we *subtract* the strong-form residual evaluated at the manufactured field,

$$F_{\text{res}} \leftarrow F_{\text{res}} - \int X|_{u_{\text{ex}}} v d\mu,$$

which makes u_{ex} the exact solution of the forced problem; U_h then converges to it at the CG1 FE rate ($L^2 \sim h^2$, $H^1 \sim h$). The sources are composed *independently* (not via `ufl.replace` on F_{res}), so any wiring bug in the residual surfaces as a convergence-rate failure instead of canceling symmetrically. Interior weak-flux terms are converted by IBP, leaving $-\text{div}(\cdot)$ on Ω ; the $\partial\Omega$ trace vanishes on the bulk (Dirichlet) and side walls ($\partial_x = 0$), so only the electrode trace remains and becomes a boundary source.

With outward normal n , ε_c = Poisson coefficient, ρ_c = charge prefactor, C_S^{model} = Stern coefficient, s_{ij} = stoichiometry, the four sources (and the term each is subtracted from) are:

1. **NP interior** (per species i): $S_{c_i} = -\text{div } J_i^{\text{ex}}$; subtract $\int_{\Omega} S_{c_i} v_i dx_q$.
2. **BV electrode** (per species i): $g_i^{\text{elec}} = J_i^{\text{ex}} \cdot n - \sum_j s_{ij} R_j^{\text{ex}}$; subtract $\int_{\Gamma_{\text{elec}}} g_i^{\text{elec}} v_i ds_q$.
3. **Poisson interior**: $S_{\varphi} = -\varepsilon_c \Delta \varphi^{\text{ex}} - \rho_c (z_H c_H^{\text{ex}} + \sum_k z_k c_k^{\text{ster,ex}})$; subtract $\int_{\Omega} S_{\varphi} w dx_q$.
4. **Stern Robin electrode**: $g_S = \varepsilon_c (\nabla \varphi^{\text{ex}} \cdot n) - C_S^{\text{model}} (\varphi_{\text{app}}^{\text{model}} - \varphi^{\text{ex}})$; subtract $\int_{\Gamma_{\text{elec}}} g_S w ds_q$.

The proton's S_{c_H} uses J_H^{ex} (the $\nabla \mu_H$ form), and R_j^{ex} in (2) carries the muh substitution $u_H \rightarrow \mu_H - e_m z_H \varphi$ inside the BV cathodic factors. Because the Stern-on branch sets $\eta_j^{\text{ex}} = \text{bv_exp_scale}(\varphi_{\text{app}}^{\text{model}} - \varphi^{\text{ex}} - E_{eq,j}^{\text{model}})$, $R_j^{\text{ex}}|_{y=0}$ is genuinely x -dependent — this is the only path that couples φ into BV, and it fires only when $C_S > 0$.

Code path. `_build_source_terms` composes (1)–(4) as UFL; `_inject_source_terms` performs the subtractions on `ctx['F_res']` and rederives the Jacobian via `fd.derivative(F, U)`:

Term	Production residual (<code>forms_logc_muh.py</code>)	Injection
(1) S_{c_i}	$\int J_i \cdot \nabla v_i dx$ (:499)	<code>F -= S_c_i * v_i * dx_q</code>
(2) g_i^{elec}	$-\sum_{ij} s_{ij} R_j v_i ds$ (:615)	<code>F -= g_elec_i * v_i * ds_q</code>
(3) S_{φ}	$\varepsilon_c \nabla \varphi \cdot \nabla w - \rho_c(\dots)$ (:644,656)	<code>F -= S_phi * w * dx_q</code>
(4) g_S	$-C_S^{\text{model}}(\varphi_{\text{app}} - \varphi) w ds$ (:668)	<code>F -= g_S * w * ds_q</code>

Quadrature is elevated to `SRC_QUAD_DEGREE = 8` on both dx_q and ds_q . Newton is initialised at $U = U_{\text{prev}} = U_{\text{manuf}}$ with $\Delta t = 10^{15}$, so the time term $(c - c_{\text{old}})/\Delta t$ is negligible and the test isolates the steady spatial operator. The convergence sweep runs at $V_{\text{RHE}} = +0.55 \text{ V}$ with $K_{0,R_{4e}}^{\text{factor}} = 10^{-18}$ (keeping $10 < \|R_{4e}^{\text{ex}}\|/\|R_{2e}^{\text{ex}}\| < 10^5$ so both reaction branches stay active). At u_{ex} all clamps/floors (`exponent_clip`, `u_clamp`, per-ion `phi_clamp`, `free_dyn_floor`, `packing_floor`) are identity, so the sources above are the unclamped/unfloored expressions.

Meshes. Rates use a uniform `UnitSquareMesh` sweep $N \in \{8, 16, 32, 64\}$; a single-solve recovery also runs on the demo's graded rectangle ($N_x = 8$, $N_y = 80$, $\beta = 3$, from `make_graded_rectangle_mesh`), which clusters nodes at the electrode by a power-law stretch of the unit-cube y -coordinate (x stays uniform):

$$y = \eta^{\beta} \hat{H}, \quad \eta \in [0, 1] \text{ uniform}, \quad \hat{H} = L_{\text{eff}}/L_{\text{ref}} (= 1 \text{ here}). \quad (5)$$

Local spacing $dy/d\eta = \beta \eta^{\beta-1} \hat{H} \rightarrow 0$ as $\eta \rightarrow 0$ for $\beta > 1$ (refinement ratio $\beta N_y^{\beta-1} \approx 1.9 \times 10^4$), so this confirms recovery of u_{ex} on the demo's actual anisotropic mesh, not just the uniform sweep.